



BITS Pilani

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CE F231

Fluid Mechanics

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Lecture 15: Fluid Kinematics

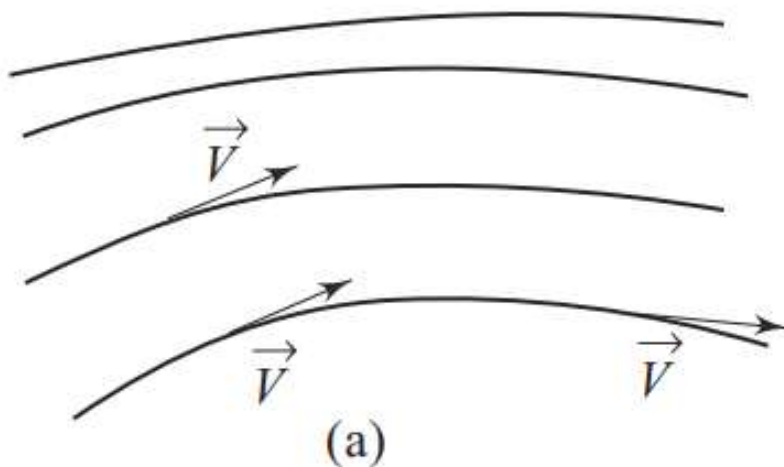
Recap: Velocity and Acceleration

- Total acceleration = Convective acceleration + Local acceleration

Total	Convective	Local
$a_x = \frac{du}{dt}$	$= u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z}$	$+ \frac{\partial u}{\partial t}$
$a_y = \frac{dv}{dt}$	$= u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z}$	$+ \frac{\partial v}{\partial t}$
$a_z = \frac{dw}{dt}$	$= u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z}$	$+ \frac{\partial w}{\partial t}$

Recap: Streamlines

- A streamline at **any time instant** can be defined as an imaginary curve in the flow field such that the tangent to the curve at any point represents the direction of instantaneous velocity at that point
- Streamlines are defined for a particular time instant (Eulerian approach)
- Two streamlines cannot intersect



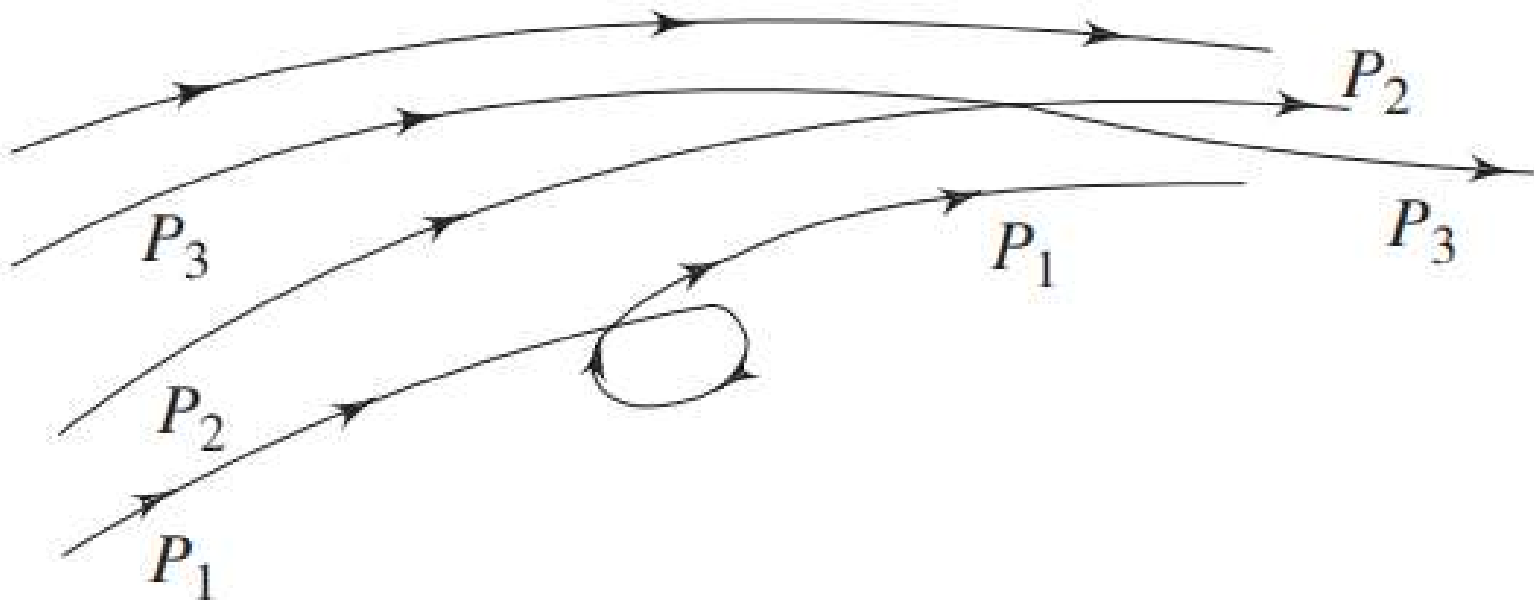
$$\vec{V} = \vec{i}u + \vec{j}v + \vec{k}w \quad \text{and} \quad d\vec{S} = \vec{i} dx + \vec{j}dy + \vec{k} dz$$

$$\vec{V} \times d\vec{S} = 0$$

$$\frac{dx}{u} = \frac{dy}{v} = \frac{dz}{w}$$

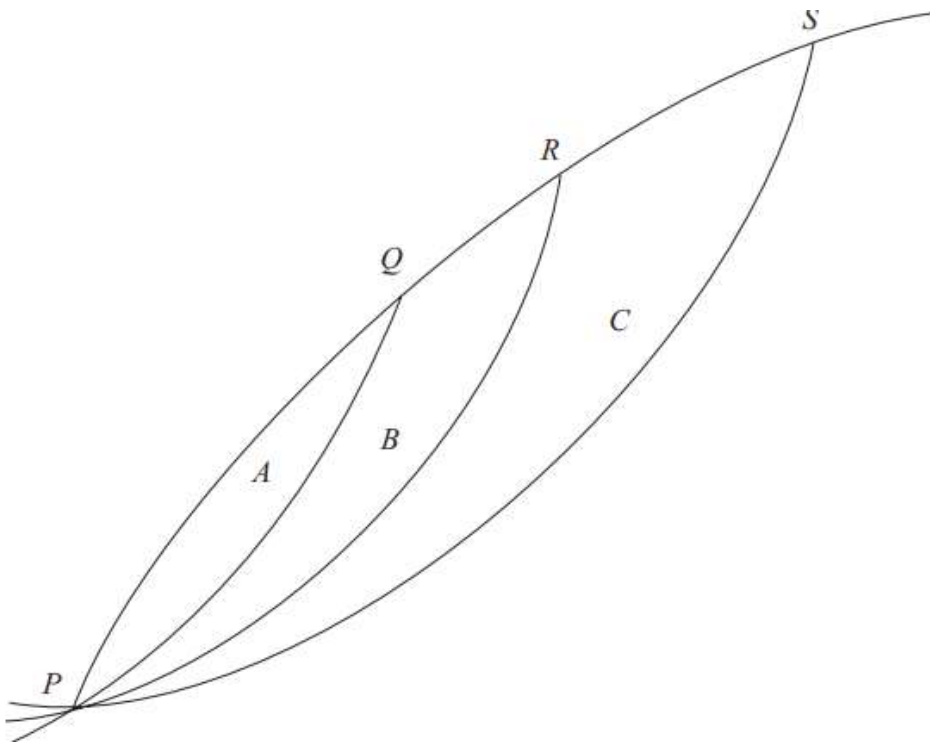
Recap: Path Lines

- Path line is the trajectory of a particular fluid particle (Lagrangian approach)
- Path lines inherently involves **variation of time**
- Two path lines can intersect, or one path line can itself form a loop



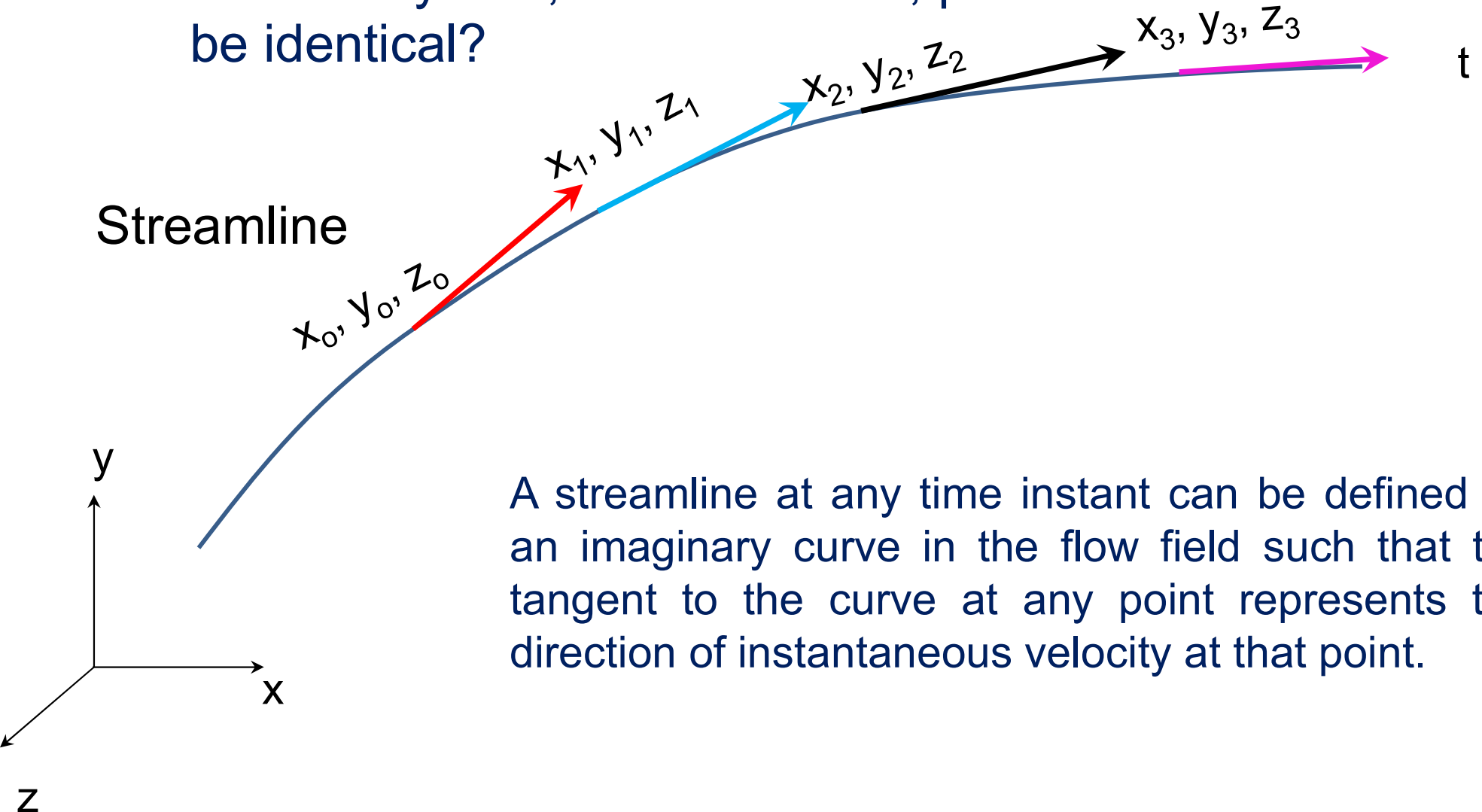
Streak Lines

- A streak line **at any instant** is the locus of the temporary locations of all particles that have passed through a fixed point in the flow field
- Streak line is important for experimental flow visualization



Question

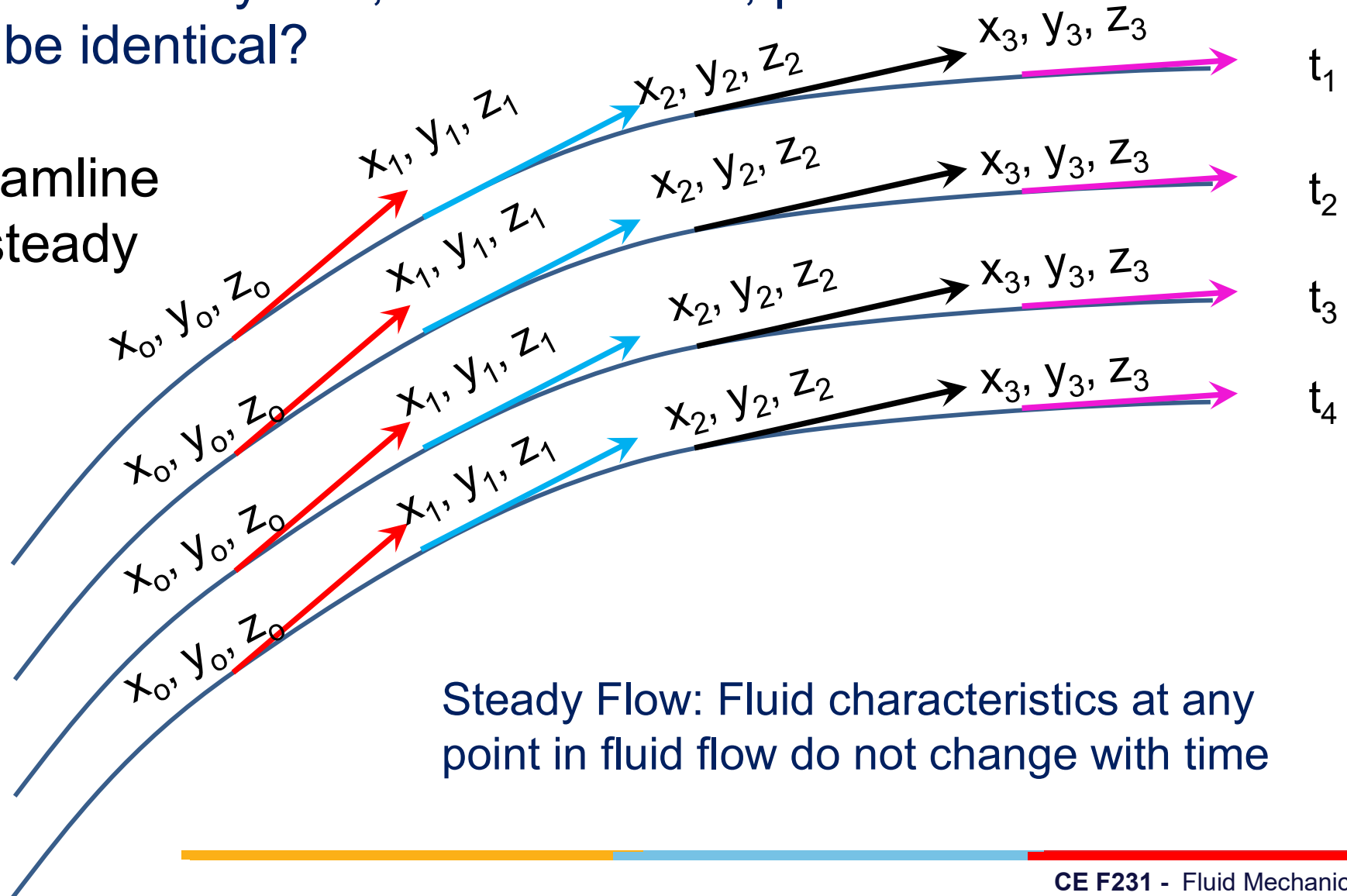
- For steady flow, will streamline, path line and streak line be identical?



Question

- For steady flow, will streamline, path line and streak line be identical?

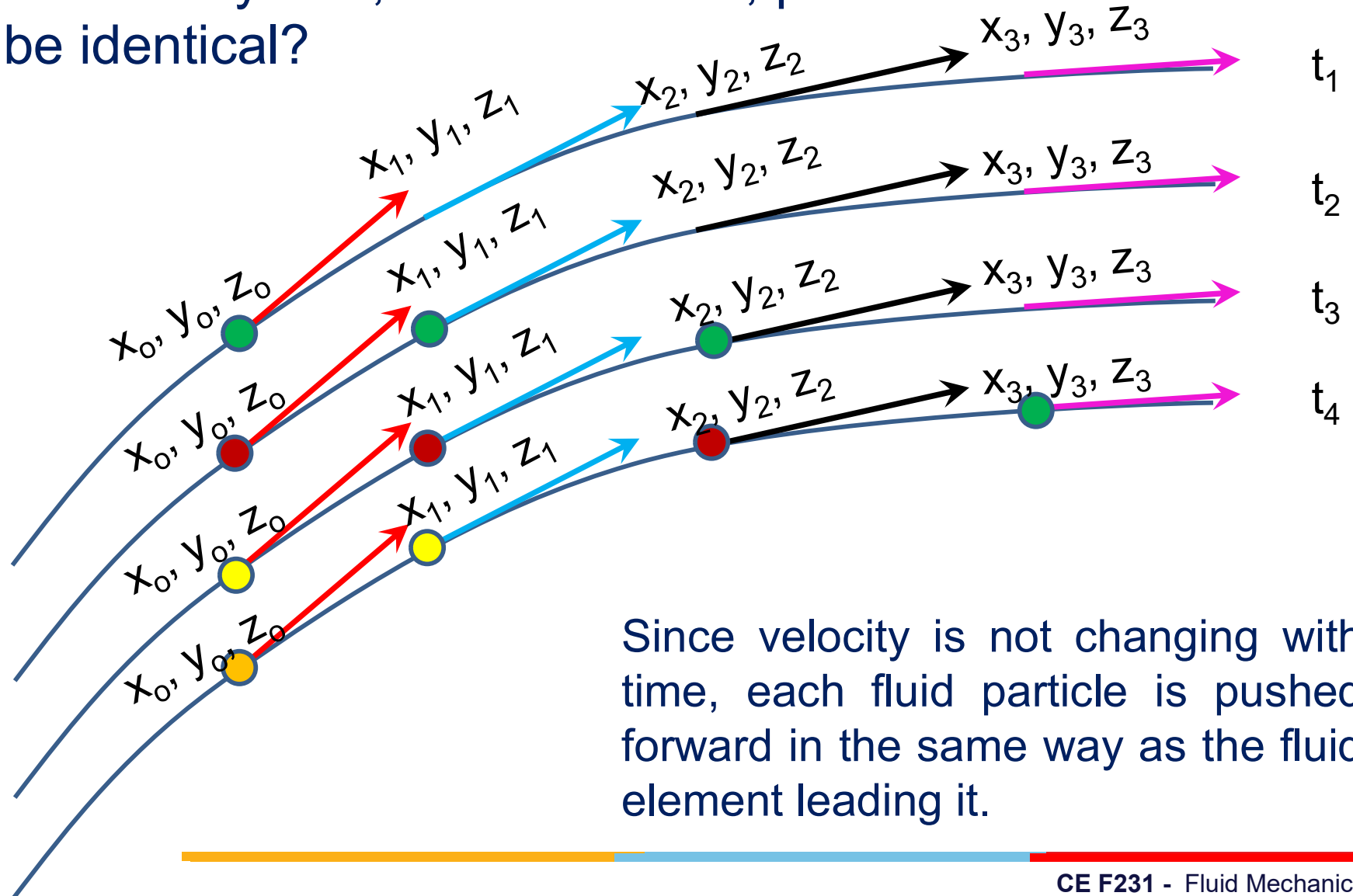
Streamline
for steady
flow



Steady Flow: Fluid characteristics at any point in fluid flow do not change with time

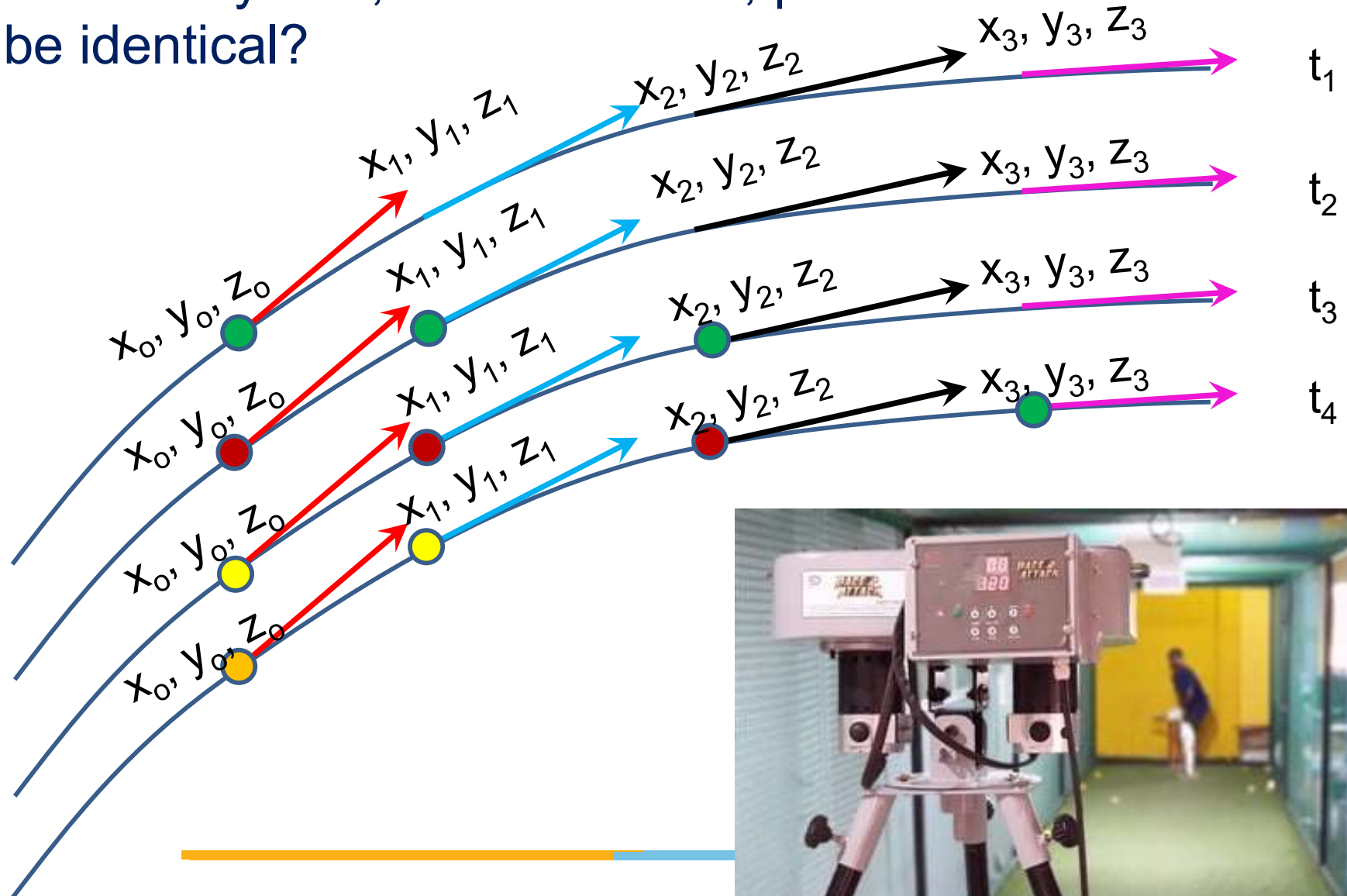
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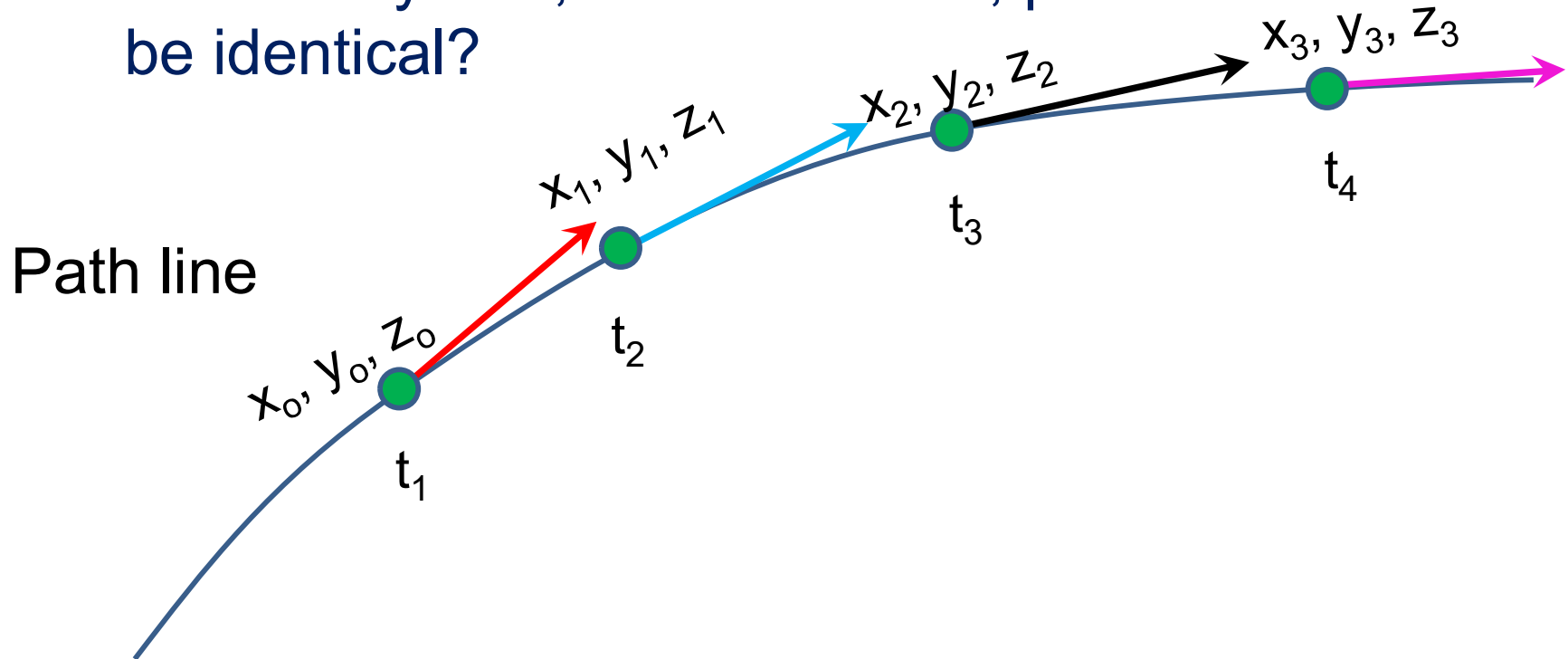
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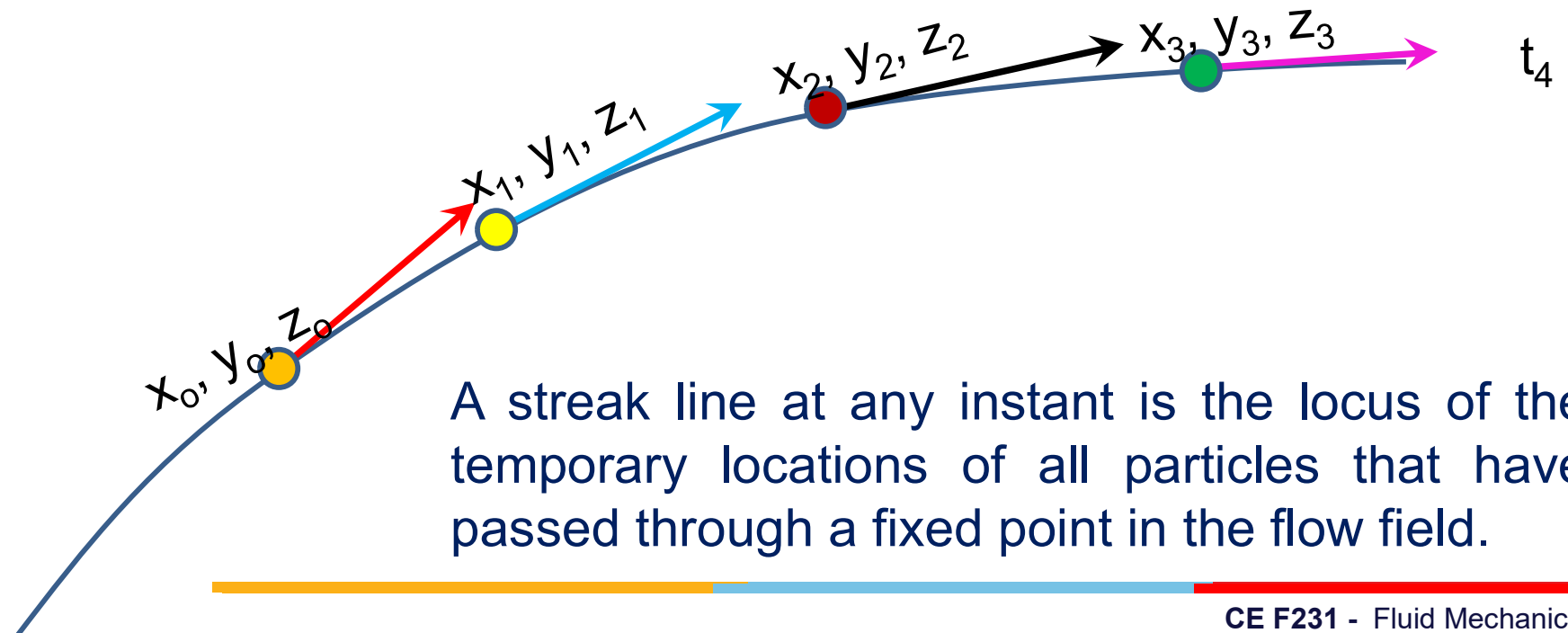


Path line is the trajectory of a particular fluid particle.
Path lines inherently involves variation of time

Question

- For steady flow, will streamline, path line and streak line be identical?

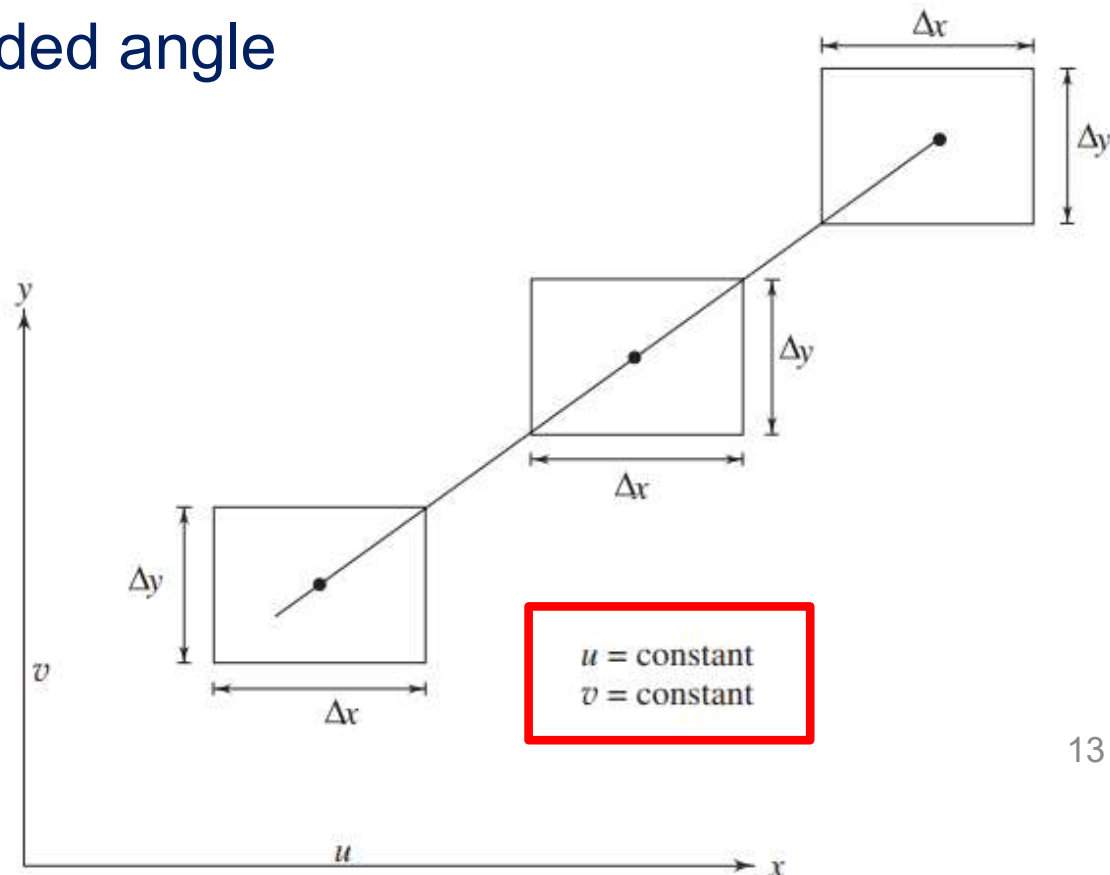
Streak line for particles passing through x_0, y_0, z_0 at time $t = t_4$



Types of Motion

➤ Linear translation

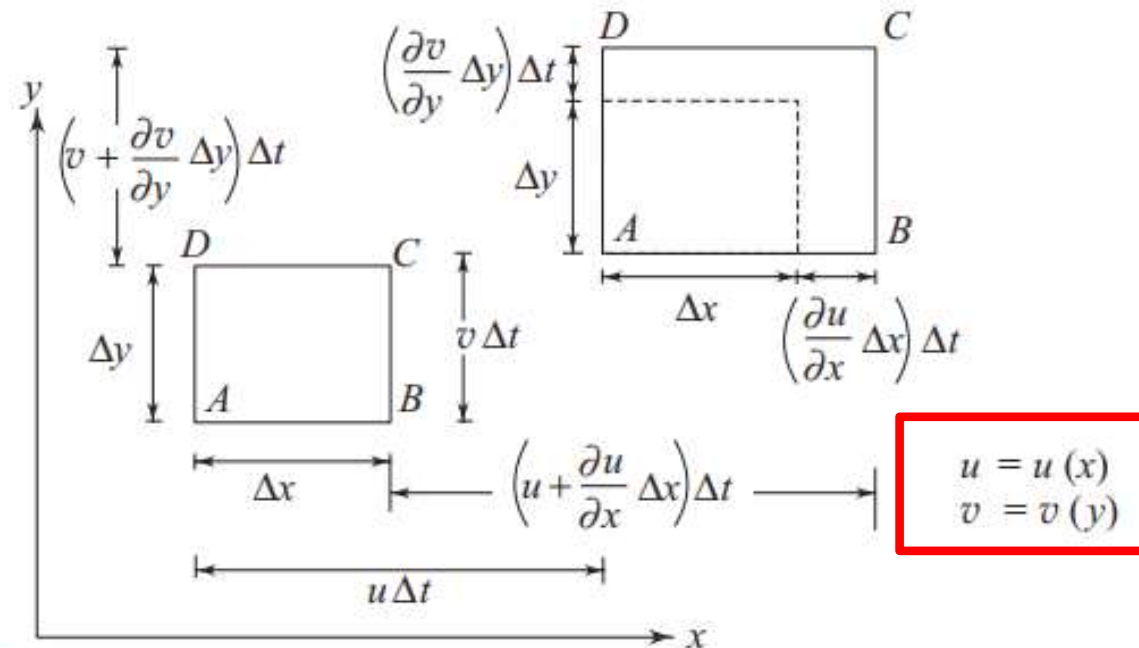
- u (x-component) and v (y-component) are not functions of x and y (space) → uniform flow
- u and v are constants
- No change in length and included angle
- Sides are displaced parallelly



Types of Motion

➤ Linear deformation

- u (x-component) and v (y-component) are functions of one space coordinate only
- u is a function of x only
- v is a function of y only
- Change in linear dimension
- No change in included angle
- Sides are displaced parallelly
- Linear strain rates



$$\dot{\epsilon}_{xx} = \frac{\partial u}{\partial x} \quad \dot{\epsilon}_{yy} = \frac{\partial v}{\partial y}$$